Hugo Marquerie 10/06/2025

Lem sigma algebras indep imp var aleatorias indep

Lema 1. *Sean $\mathcal{F}_1, \mathcal{F}_2 \subset \Sigma$ dos σ-álgebras independientes en $(\Omega, \Sigma, \mathbb{P})$ y $X_1 \colon \mathcal{F}_1 \to \mathbb{R}$, $X_2 \colon \mathcal{F}_2 \to \mathbb{R}$ dos variables aleatorias $\implies$ X_1 y X_2 son independientes.*

***Demostración*:** Tenemos que $\forall i = 1, 2 : \forall E_i \in \mathcal{B}(\mathbb{R}) : \{X_i \in E_i\} \in \mathcal{F}_i$ y, como $\mathcal{F}_1$ y $\mathcal{F}_2$ son independientes, $\forall A_1 \in \mathcal{F}_1, A_2 \in \mathcal{F}_2 : \mathbb{P}\left(A_1 \cap A_2\right) = \mathbb{P}(A_1) \cdot \mathbb{P}(B_2)$.

$$\implies \mathbb{P}(A_1 \cap A_2) = \mathbb{P}\left(\{X \in E_1\} \cap \{X \in E_2\}\right) = \mathbb{P}\left(X \in E_1\right) \cdot \mathbb{P}\left(Y \in E_2\right) = \mathbb{P}(A_1) \cdot \mathbb{P}(A_2)$$

donde $A_i = \{X_i \in E_i\}$. Como A_1 y A_2 eran arbitrarios, X_1 y X_2 son independientes. ■

Temporary page!

LaTeX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because LaTeX now knows how many pages to expect for this document.